Joshua Niemelä

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Education

Copenhagen University (DIKU) **08/2022 →**
Bachelor of Science in Machine Learning and Data Science
Wrote various machine learning models to attempt to classify news articles by trustworthiness for a school project.

Niels Brock (DIG) **08/2019 – 06/2022**
Higher Commercial Exam (HHX) w. supplementary physics, chemistry and math.

Professional experience

Promilist **01/2023 – 03/2023**
Freelance Developer

- Migrated Python codebase to Julia. Improved performance and maintainability by refactoring to a more pure and modularised codebase.
- Wrote code to calculate fuel-optimal paths for shipping vessels by traversing space-time graphs on a geodesic map.
- Implemented a Genie.jl API to interface with the existing backend.

Relevant skills:

- Performance testing
- Distributed computing
- GPU programming
- Shortest path optimisation

Juristic ApS **08/2022 – 02/2023**
Machine Learning Developer
Developed an API and pipeline in Julia from start to finish, which utilised morphological image processing, Google Vision AI and machine learning to convert handwritten relational charts on whiteboards into a digital data format. Implemented CI/CD for scalable deployment on Scalingo.

Relevant skills:

- Morphological image processing
- Unsupervised machine learning / Clustering
- Image recognition
- Test-driven development

Skills

Programming languages:

Julia: High performance parallel computing language. Wrote a small package, Vision.jl, for interfacing with Google's Vision API in Julia.

Python: Experience with pandas, NumPy, scikit-learn, OpenCV and TensorFlow. Written various programs for image segmentation, semantic text classification and time series analysis.

Haskell: Wrote a clone of the game 2048 with a TCP socket to communicate with other programs. Wrote an interpreted dialect of Scheme with fexprs and vau-calculus in place of macros.

Clojure: Wrote a backend to interact with a Neo4j graph database for a note-taking/zettelkasten application, currently writing a corresponding frontend in ClojureScript.

Miscellaneous:

Linux: Daily linux user and acquainted to a CLI-based workflow.

Docker/Nix: Experienced with containerisation and using Nix/Docker for continuous integration and ensuring reproducible environments for code in production.

Languages

English/Danish: Native/C1

German: Conversational